ALICE L WALTON SCHOOL OF MEDICINE

Cancer Medications Overview

Last updated October 20, 2025 9 min read CC BY-SA 4.0

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Adapted from: Cancer Therapy: Foundations and Frameworks

Listen to this Brick
0:00 / 15:44 1x

Learning Objectives (2) Versions

After completing this brick, you will be able to:

1. Compare and contrast mechanisms of action of drugs commonly used in cancer therapy.
2. Recognize common adverse effects of drugs commonly used in cancer therapy.

CASE CONNECTION

LM is a 7-year-old boy recently diagnosed with acute lymphoblastic leukemia (ALL). He has started induction chemotherapy, which includes vincristine as part of his regimen. After several weeks of treatment, LM's parents bring him to the oncology clinic with concerns about new symptoms. You are a student rotating through the oncology clinic. The parents report that he has been experiencing difficulty walking, frequent tripping, and complains of tingling and numbness in his fingers and toes. His physical exam reveals decreased deep tendon reflexes in the lower extremities, mild foot drop, and reduced grip strength.

LM's parents ask, "How does this vincristine work anyway? Are these the normal side effects of this drug?" What will you tell LM's parents? Consider your answer as you read, and we'll revisit LM at the end of the brick.

GO TO CONCLUSION ↓

What Is Cancer Pharmacotherapy?

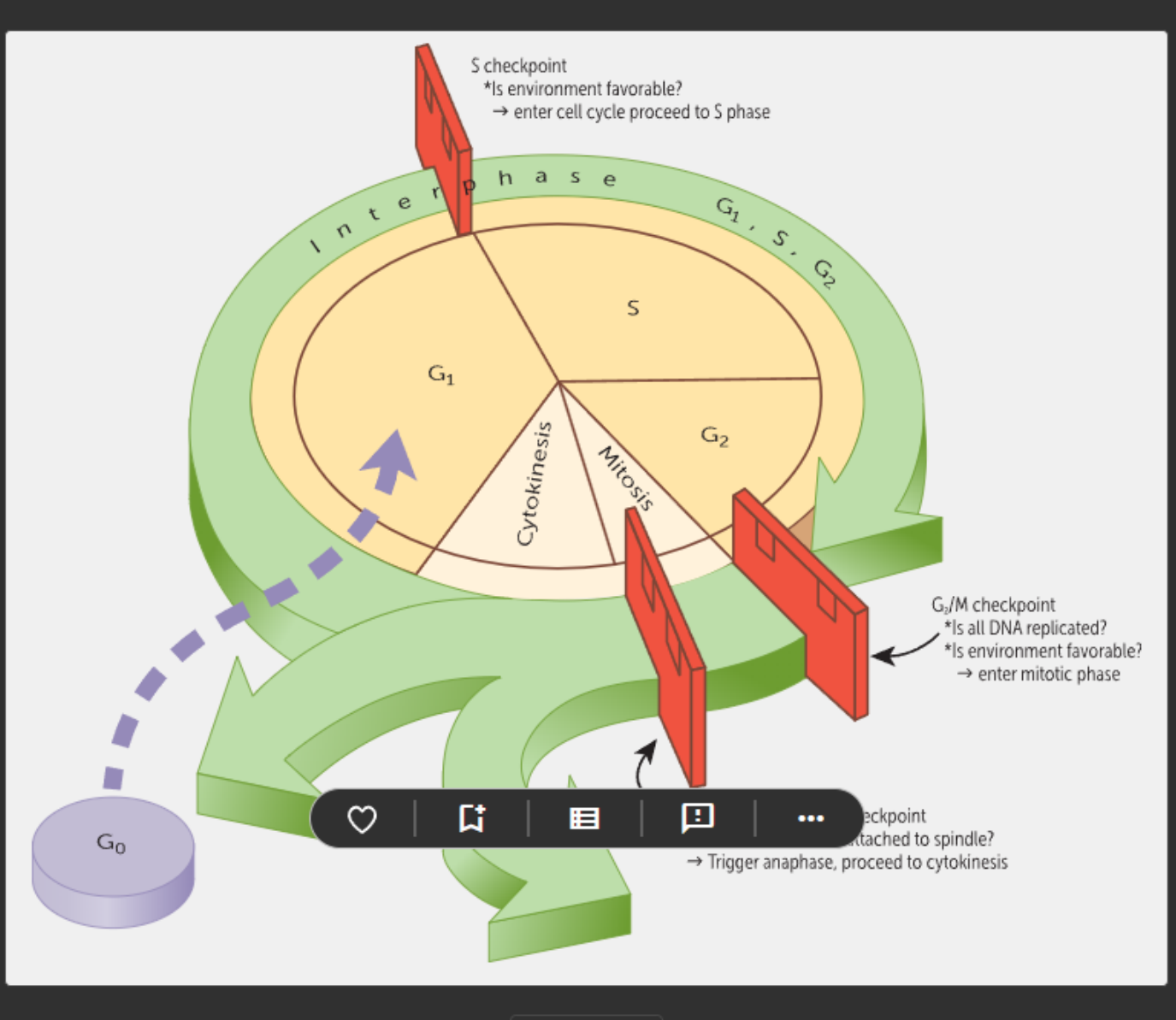
Cancer treatment can be broadly classified into three main categories: surgery, radiation therapy, and medical treatment. Medical treatment, or pharmacotherapy, includes (but is not limited to) traditional cytotoxic chemotherapy, molecularly targeted therapy, immunotherapy, and hormone (endocrine) therapy. In many instances, combination therapy is used. Immunotherapies will be discussed briefly below and covered in depth in a later course. Hormone therapies, like tamoxifen a selective estrogen receptor modulator (SERM) used in estrogen receptor (ER) positive breast cancer, will be discussed in context i rposes of this brick we will primarily focus on the traditional cytotoxic chemotherapy agents.

How Does Chemotherapy Disrupt Cellular Pathways? (LO#1)

To explore these mechanisms, a review of the cell cycle will help.

Cell Cycle

The cell cycle describes the steps by which a cell divides. Figure 1 illustrates the mitotic cell cycle for a typical human cell.



QUIZ

Figure 1

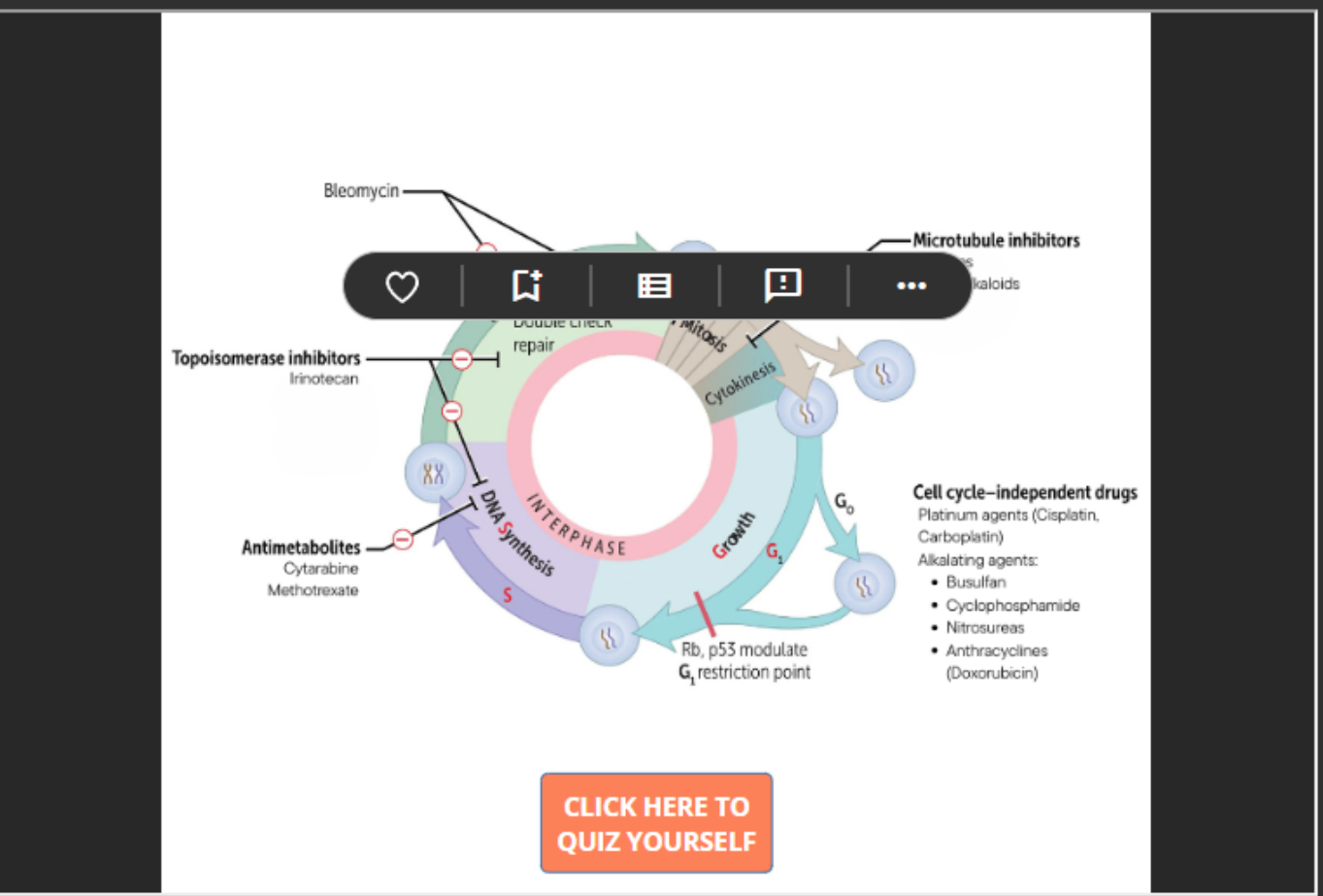
Nondividing cells are in the G_0 phase. Some cells, like those in skeletal muscle and the brain, stay more or less permanently in the G_0 phase, whereas others, like those in the epidermis, will regularly enter the cell cycle in interphase. **Interphase** refers to the time that cells do not spend in mitosis and includes the G_1, S, and G_2 phases. In the first gap (G_1) phase, the cell synthesizes messenger RNA and proteins required for the remainder of the cycle. In the synthesis (S) phase, chromosomal DNA is replicated in the cell. In the second gap (G_2) phase, the cell synthesizes more protein, including microtubules, which are required for the mitotic spindle. Then, the cell undergoes mitosis (M) and repeats the cycle.

Along the cell cycle, a series of **cell cycle checkpoints** ensures that the cell is truly ready to replicate. Any damage to the DNA or an unfavorable environment halts the cell cycle and causes the cell to undergo **apoptosis** (programmed cell death). In this way, the body can get rid of "bad" cells that contain damaged DNA. In cancerous cells, however, many of these checkpoints are broken, allowing uncontrolled division of such bad cells. The three main checkpoints in the cell cycle are the G_1/S, G_2/M, and metaphase/anaphase checkpoints.

What is the purpose of cell cycle checkpoints?

Mechanisms of Action of Cytotoxic Chemotherapy Drugs

Cytotoxic chemotherapy drugs interrupt the cell cycle and interfere with cell division, using several mechanisms. They are classified according to the phase of the mitotic cell cycle they inhibit (Figure 2). This is not a comprehensive list, but provides some of the most common classes, and some specific agents, that you should be aware of.



S Phase: Antimetabolites. All of the medications in this class have slightly different mechanisms; however, all of them are specific to the S phase and prevent the cell from progressing from G_1 to S phase.

Drugs like methotrexate inhibit DNA synthesis (S phase), repair, and cellular replication. Methotrexate binds to and inhibits dihydrofolate reductase, inhibiting the formation of reduced folates, and thymidylate synthetase, resulting in inhibition of purine and thymidylic acid synthesis, thus interfering with DNA synthesis, repair, and cellular replication.

Cytarabine is a pyrimidine analog and is incorporated into DNA; however, the primary action is inhibition of DNA polymerase resulting in decreased DNA synthesis and repair.

S/G Phases: Topoisomerase Inhibitors. Drugs in this class disrupt the action of topoisomerases. Irinotecan and its active metabolite bind reversibly to topoisomerase I-DNA complex preventing relegation of the cleaved DNA strand. This results i lexes and double-strand DNA breaks. As cells cannot efficiently repair these breaks, cell death consistent with S-phase cell cycle specificity occurs, leading to termination of cellular replication.

G_2 Phase. Bleomycin inhibits synthesis of DNA, binds to DNA leading to single- and double-strand breaks, and also inhibits (to a lesser degree) RNA and protein synthesis.

M Phase: Microtubule Inhibitors. The **vinca alkaloids** bind to tubulin and inhibit microtubule formation, therefore, arresting the cell at metaphase by disrupting the formation of the mitotic spindle; it is specific for the M phase.

Taxanes promote microtubule assembly by enhancing the action of tubulin dimers, stabilizing existing microtubules, and inhibiting their disassembly, interfering with mitotic phase, and inhibiting cell replication.

Cell Cycle–Independent Drugs: The **alkylating Agents,** such as busulfan, cyclophosphamide, doxorubicin (anthracycline class), and nitrosureas, operate independently of the cell cycle. They cross-link DNA chains and cause them to break, inhibiting DNA replication and transcription. This same mechanism is observed in the **platinum agents,** carboplatin and cisplatin. These classes of chemotherapy drugs are used in may different types of cancer including, breast, bladder, leukemias/ lymphomas, lung cancers, and many others.

How do alkylating agents inhibit DNA replication?

Molecularly Targeted Therapy & Immunotherapy

A number of newer monoclonal antibodies target specific cell receptors or structures that are key in promoting cancer growth. Three classic targeted agents include rituximab, bevacizumab, and trastuzumab. **Rituximab** acts on CD20 receptors on B cells, making them more susceptible to attack by the immune system. It is used for B-cell malignancies like non-Hodgkin lymphoma and chronic lymphocytic leukemia. **Bevacizumab** binds to vascular endothelial growth factor (VEGF) in blood vessels to inhibit the angiogenesis (abnormal blood vessel proliferation) that accompanies tumor growth. Without the vascular supply, the cancer cells die. **Trastuzumab** binds to HER2 receptors on tumor cells, blocking signaling pathways that drive growth and marking the cells for immune-mediated destruction. It is primarily used for HER2-positive breast cancer

CLINICAL CORRELATION

Any drug that ends in the suffix -mab is a therapeutic monoclonal antibody, which targets specific cell

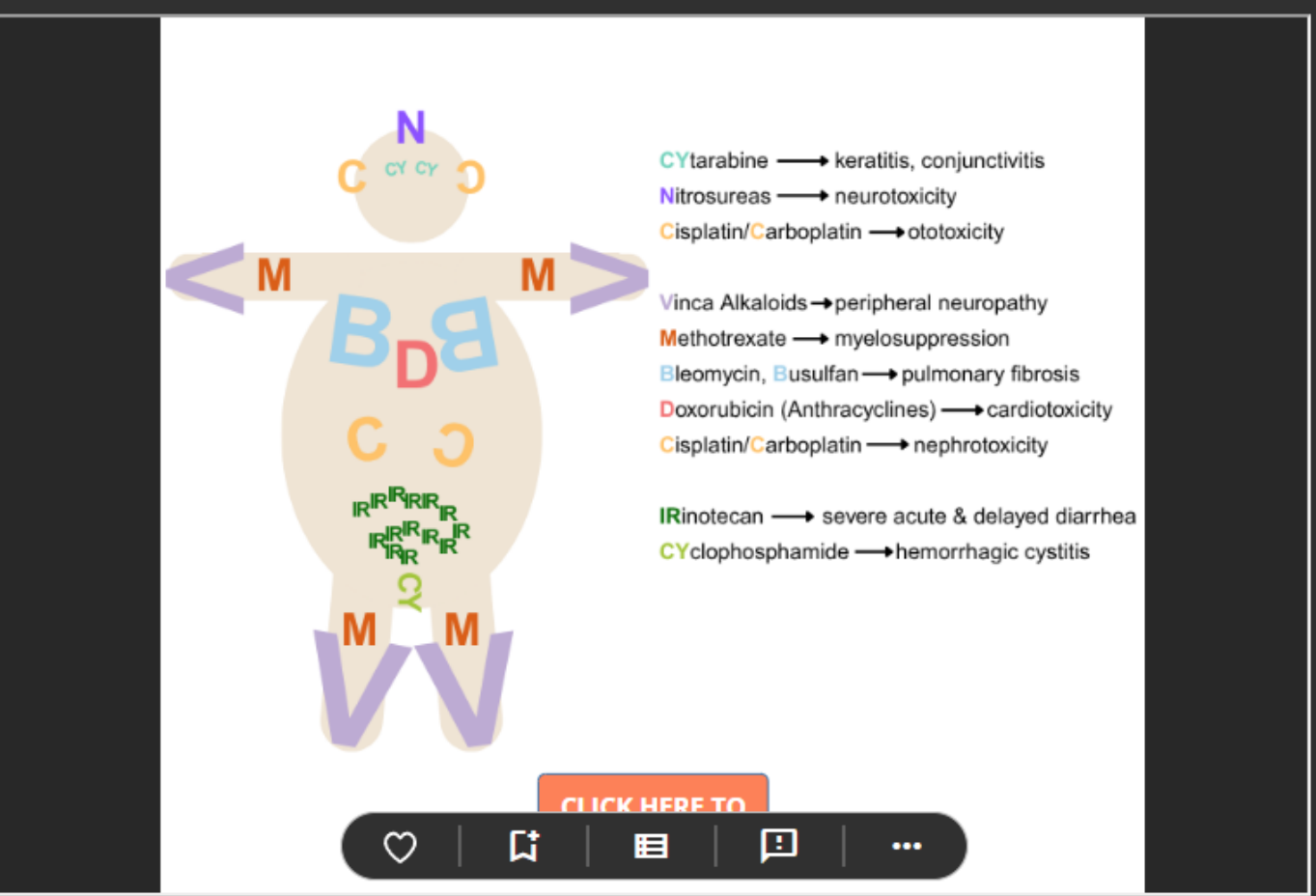
Immunotherapy is the latest cancer treatment to be rolled out. With immunotherapy, the body's own immune system is used to fight off cancer. The immune system plays an essential role in recognizing and eliminating tumor cells, which, after all, are "foreign" and should stimulate an immune response. However, cancer cells can use inhibitory signaling pathways called **immune checkpoints** to "turn off" the antitumor immune response. Many drugs will target these areas to exert their effects.

Cell Therapy is another, newer, cancer therapy. This includes a newer kind of immunotherapy called **chimeric antigen-receptor T cells (CAR-T).** This is a cell therapy rather than a drug. CAR-T cells are genetically modified autologous (belonging to the patient) T cells. The cells are altered by taking the antigen-binding domain from a B-cell receptor specific for a cancer antigen, then fusing it onto the patient's T-cell CD3 receptor. This creates a "chimeric" receptor (meaning it has structures from different cell origins). Then, the modified T cells are injected back into the patient. The modified T cells now have a specific receptor for the tumor cells, making it much easier for the T cells to target the cancer.

Molecularly targeted, immunotherapies, and cell therapy to treat cancer will be discussed more in the coming courses. As mentioned above, this brick is meant to introduce cytotoxic chemotherapy drugs, but an awareness of the other modalities is helpful.

What Are the Adverse Effects of Chemotherapeutic Drugs? (LO#2)

Because most chemotherapeutic drugs target any rapidly dividing cell, not just cancer cells, they also lead to the death of normal cells that turn over rapidly. Based on what you learned about the mechanisms of these drugs in the last section of this brick, you can imagine that they would have some significant adverse effect profiles. Each of the drugs will have varying affinities for rapidly dividing cells throughout the body. So, drugs that affect the gastric mucosa will likely lead to nausea and vomiting, drugs that affect bone marrow cells will result in myelosuppression, etc. Some of the most common classes and agents are compiled into one image, named ChemoMan (Figure 3). This is an important image to become very familiar with! You will see it again in future courses, board exams, etc.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C CY CY C
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CYtarabine → keratitis, conjunctivitis
Nitrosureas → neurotoxicity
Cisplatin/Carboplatin → ototoxicity
Vinca Alkaloids → peripheral neuropathy
Methotrexate → myelosuppression
Bleomycin, Busulfan → pulmonary fibrosis
Doxorubicin (Anthracyclines) → cardiotoxicity
Cisplatin/Carboplatin → nephrotoxicity
IRinotecan → severe acute & delayed diarrhea
CYclophosphamide → hemorrhagic cystitis
CLICK HERE TO

The more specific agents like monoclonal antibodies react against the cancer cells in a more targeted way, which reduces the number and intensity of adverse effects. Each of those will be covered as they are discussed in future courses.

CASE CONNECTION

BACK TO INTRODUCTION ↑

Thinking back to LM, how does vincristine work? Is this a normal adverse effect? As you can imagine, the parents are concerned about their child having to going through chemotherapy. Expressing concern and answering their questions will be very important. You might briefly explain that cancer occurs when cells in the body rapidly grow and spread uncontrolled. Vincristine works by inhibiting microtubules, which arrests cells in M phase not allowing replication, growth, or spread. This is effective at killing cancer cells, but it can also effect other cells that grow rapidly. Vincristine in particular can effect cells in the nerves that cause the peripheral neuropathy, or symptoms of tingling and numbness that is causing him to trip and fall. Since they reported it early it can be managed with reducing doses and continued monitoring.

Summary

- Cancer pharmacotherapy includes surgery, radiation therapy, and medical treatment, with medical treatment encompassing cytotoxic chemotherapy, molecularly targeted therapy, immunotherapy, and hormone therapy.
- Cell cycle checkp aphase) are crucial for halting replication of cells with damaged DNA, but these checkpoints are often broken in cancer cells, leading to uncontrolled division.
- Cytotoxic chemotherapy drugs disrupt the cell cycle and interfere with cell division by targeting specific phases, such as antimetabolites for the S phase and microtubule inhibitors for the M phase.
- Adverse effects of chemotherapeutic drugs arise from their action on rapidly dividing cells, affecting normal cells and leading to side effects like myelosuppression and cardiotoxicity.

Review Questions

1. **Which of the following cell cycle phases is targeted by antimetabolites such as methotrexate?**

   A. ○ G_0
   B. ○ G_1
   C. ○ G_2
   D. ○ M
   E. ● S

Hide Explanation

The correct answer is oprine and methotrexate inhibit dihydrofolate reductase enzyme. This enzyme is necessary for DNA synthesis, which occurs during the S phase. Anticancer drugs do not target the G_0 (A) phase; the cell does not divide in this phase, so it has minimal malignant potential. The G_1 (B), G_2 (C), and M (D) phases are not targeted by antimetabolites.

2. **Which chemotherapy agent is associated with cardiotoxicity?**

   A. ○ Vinca Alkaloids
   B. ○ Cytarabine
   C. ● Doxorubicin
   D. ○ Methotrexate
   E. ○ Irinotecan

Hide Explanation

The correct answer is Doxorubicin (Anthracyclines) (C). Doxorubicin is known for its cardiotoxic effects. Vinca Alkaloids (A) are associated with neurotoxicity. Cytarabine (B) can cause keratitis and conjunctivitis. Methotrexate (D) is known for causing myelosuppression. Irinotecan (E) is associated with severe diarrhea.

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How would you rate this Brick?

Review Learning Objectives

Check off the objectives when you feel comfortable with the concept

01 ☑ ~~Compare and contrast mechanisms of action of drugs commonly used in cancer therapy.~~

02 ☑ ~~Recognize common adverse effects of drugs commonly used in cancer therapy.~~

Objective Confidence: 2/2